Single-channel EEG seizure detection on CHB-MIT: accuracy, model size and efficiency

Bold = best in column. Underlined = second best. Higher is better for ACC and SEN; lower is better for parameters, stored footprint and MACs. Rows on a dataset other than CHB-MIT are shown for reference and excluded from the ranking.

Method	Data	Model	Protocol / test exposure	ACC (%)	SEN (%)	Parameters	Precision	Stored model footprint	Other efficiency
WearSeizure-1D (this work)	CHB-MIT (13 single-channel cases)	Separable 1D-CNN, k5 + dilation 1/2/4/8/16; 1 channel; 4-s windows	**Leakage-safe**: split by recording before filtering; thresholds frozen on val; **185.0 h** continuous test; 66 folds x 3 seeds	98.88	94.89 event level; 60.33 segment level	11,786	INT8 (target)	11.5 KB INT8 weights; 18.2 KB total on-chip SRAM (weights + line buffers)	585,920 MACs (thop) / 489,600 conv+fc; 6.7 KB line buffers; FAR 0.29/h; delay 17.8 s
WearSeizure-1D, same code under the segment-split protocol	CHB-MIT (13 single-channel cases)	As above, retrained under the leaky protocol	*Segment-level random split*; normalised on all data; threshold on test — i.e. the protocol common to the rows below. 99.6% of test windows have a near-duplicate in training	**99.68**	92.29 segment level	11,786	FP32 (measured)	as above	Same network; only the split rule differs from the row above
Chung et al. 2024 [1]	CHB-MIT (13 cases)	Stacked 2D-CNN, parallel 1x3 and 1x5 kernels; 1 clinician-selected channel	Segment-level 70/20/10 split per patient; ~91 h	98.18	**99.62** event level; 96.76 segment level	116,700	FP32	NR; approximately 467 KB if all weights are FP32	FAR 0.22/h; detection delay 3.3 s; specificity 98.19%
EpiSepNet-5K (FP32)	CHB-MIT	Separable 1D-CNN; 17-channel raw EEG; 2-s windows	NR	90.07	90.76	5,010	FP32	28.1 KB checkpoint; approximately 20.0 KB raw FP32 weights	Reference model
EpiSepNet-5K (INT16)	CHB-MIT	BatchNorm-folded separable 1D-CNN	NR	90.04	90.76	**4,900**	INT16	10.0 KB package; approximately 9.8 KB raw INT16 values	99.9743% agreement with FP32; 2.81x smaller package
Werner et al. (TC-ResNet4) [2]	CHB-MIT	16-channel TC-ResNet4 with fixed-point inference	NR	95.28	92.34	9,840	4-bit	**Approximately 4.92 KB weight-only storage; packaged size NR**	**337,968 MACs; 495 nW average power**
Zhu et al. 2021 [3]	CHB-MIT (23 channels)	7-layer 1D-CNN, 5x1 conv, shift-register PE array	80/10/10 split; normal class down-sampled 200:1	97.35	94.32	7,010	fixed-point	NR	6.32 MOPs; 170 us/inference @ 200 MHz; FPGA Xilinx Zynq ZC706
Li et al. 2022 [4]	CHB-MIT + Bonn + SWEC-ETHZ	1D-CNN with parallel convolutional layers	80/20 + 5-fold CV; GAN-synthesised preictal segments	99.01	99.24	10,778	analog RRAM	NR	1.13 us parallelised; 7.21 W; ASIC 22 nm FDSOI + RRAM crossbar
Ferrara et al. [5]	CHB-MIT	Patient-specific two-channel lightweight CNN	NR	99.00	67.00	9,500	NR	51 KB model	Balanced accuracy 83.0%; 0.10 FP/h
REST-RS [6]	CHB-MIT	Graph-based residual state-update model	NR	NR	NR	9,300	NR	0.037 MB, approximately 37 KB	AUROC up to 93.5%; 1.314 ms inference
SlimSeiz [7]	CHB-MIT	Eight-channel convolution and Mamba network (prediction task)	NR	94.80	95.50	21,200	NR	NR; approximately 84.8 KB if all weights are FP32	Specificity 94.0%
RGF-Model [8]	CHB-MIT	Multi-teacher knowledge-distillation model	NR	98.92	98.54	82,000	FP32	0.33 MB	Specificity 99.11%; AUC 98.96%
Wang et al. (MSCA) [9]	CHB-MIT	Inverted residual CNN with multi-scale channel attention	NR	98.70	98.30	88,000	NR	NR; approximately 352 KB if all weights are FP32	2.68M MACs; specificity 99.10%
Ahlawat (INT8) [10]	CHB-MIT	Quantized common-channel 1D-CNN with operator fusion	NR	NR	NR	NR	INT8	0.44 MB model	Up to 2.8x speedup; up to 64% estimated energy reduction
Lee et al. 2024 (RVDLAHA) [11]	*Rat skull EEG (not CHB-MIT)*	1D-CNN with RISC-V DLA; on-device personalisation	64/16/20 split	99.00	99.00	356	fixed-point	NR	2.1 ms; 107 mW @ 1 MHz; FPGA Xilinx PYNQ-Z2

Reading the table. Protocols differ between rows, and most papers do not state theirs in enough detail to compare. This work is the only row evaluated with splits taken by recording, thresholds frozen on a held-out validation partition, and 185 h of continuous test exposure. Under the segment-level random split used by much of this literature, the same code and data reach 92.29 % segment sensitivity instead of 60.33 % — a 31-point difference caused by the protocol alone, because 99.6 % of test windows then have a near-duplicate in training. Accuracy is nearly blind to this: at 0.62 % ictal prevalence a model that never predicts a seizure already scores 99.38 %.

References

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[6] REST-RS: graph-based residual state-update model for EEG seizure detection.
[7] SlimSeiz: convolution and Mamba network for seizure prediction.
[8] RGF-Model: multi-teacher knowledge distillation for seizure detection.
[9] Wang et al. Lightweight Seizure Detection Based on Multi-Scale Channel Attention, 2023.
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